

Speech Signal Processing

BSAI – 7th Semester

Open Ended Lab Task Report

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Objective

The objective of this open-ended lab is to design and implement a complete speech enhancement system in MATLAB. The system performs voice activity detection (VAD), noise estimation, spectral subtraction, cepstrum-based pitch detection, bandpass filtering, and quality analysis using SNR and visual inspection.

Task 1: Noisy Speech Signal Acquisition

MATLAB Code:

```
1 clc;
2 clear;
3 close all;
4 [x, fs] = audioread('D:\matlab-jd\speech1.wav');
5 x = x(:,1);
6 disp('Done');
```

Explanation: A real-world noisy speech signal is loaded using `audioread`. Only one channel is selected in case of stereo input. This signal serves as the input for all subsequent processing stages.

Task 2: Framing of Speech Signal

MATLAB Code:

```
1 frameLen = round(0.02*fs);
2 frameShift = round(0.01*fs);
3
4 frames = buffer(x, frameLen, frameLen-frameShift, 'nodelay');
5 numFrames = size(frames,2);
```

Explanation: The speech signal is divided into overlapping frames of 20 ms with a 10 ms shift. Framing allows short-time analysis, which is essential for speech processing because speech is quasi-stationary over short durations.

Task 3: Voice Activity Detection using Energy and Cepstrum

MATLAB Code:

```
1 energy = sum(frames.^2);
2 energy = energy / max(energy);
3 vad = zeros(1,numFrames);
4
5 for i = 1:numFrames
6     X = fft(frames(:,i));
7     cep = real(ifft(log(abs(X)+eps)));
8     [~, q] = max(cep(30:200));
9
10    if energy(i) > 0.1 && q > 0
11        vad(i) = 1;
```

```

12     end
13 end

```

Explanation: Voice Activity Detection (VAD) is performed using short-time energy and cepstral peak detection. Frames with sufficient energy and a valid pitch-related cepstral peak are classified as voiced speech, while others are treated as noise-only frames.

Task 4: Noise Spectrum Estimation

MATLAB Code:

```

1 noiseFrames = frames(:, vad == 0);
2 if isempty(noiseFrames)
3     noiseFrames = frames(:,1:5);
4 end
5
6 noiseSpec = mean(abs(fft(noiseFrames)), 2);

```

Explanation: Noise is estimated from unvoiced frames detected by VAD. The average magnitude spectrum of these frames is computed and used as the noise estimate for spectral subtraction.

Task 5: Spectral Subtraction Based Enhancement

MATLAB Code:

```

1 enhancedFrames = zeros(size(frames));
2
3 for i = 1:numFrames
4     Y = fft(frames(:,i));
5     magY = abs(Y);
6     phaseY = angle(Y);
7
8     magClean = magY - noiseSpec;
9     magClean(magClean < 0) = 0;
10
11     enhancedFrames(:,i) = real(ifft(magClean .* exp(1j*phaseY)));
12 end

```

Explanation: Spectral subtraction is applied by subtracting the estimated noise spectrum from each frame's magnitude spectrum while preserving the phase. Negative values are clipped to zero to avoid distortion.

Task 6: Signal Reconstruction using Overlap-Add

MATLAB Code:

```

1 signalLength = frameShift*(numFrames-1) + frameLen;
2 enhanced = zeros(signalLength,1);
3
4 idx = 1;
5 for i = 1:numFrames
6     enhanced(idx:idx+frameLen-1) = ...
7         enhanced(idx:idx+frameLen-1) + enhancedFrames(:,i);

```

```

8     idx = idx + frameShift;
9 end
10
11 enhanced = enhanced / max(abs(enhanced));

```

Explanation: The enhanced frames are combined using the overlap-add method to reconstruct the time-domain enhanced speech signal. Normalization prevents amplitude clipping.

Task 7: Bandpass Filtering

MATLAB Code:

```

1 bpFilt = designfilt('bandpassfir', ...
2     'FilterOrder',100, ...
3     'CutoffFrequency1',300, ...
4     'CutoffFrequency2',3400, ...
5     'SampleRate',fs);
6
7 x_filt = filter(bpFilt, x);
8 enhanced_filt = filter(bpFilt, enhanced);

```

Explanation: A bandpass filter (300–3400 Hz) is applied to retain only the speech frequency range, improving clarity and removing unwanted low- and high-frequency noise.

Task 8: Cepstrum-Based Pitch Detection

MATLAB Code:

```

1 pitch = zeros(1,numFrames);
2
3 for i = 1:numFrames
4     X = fft(frames(:,i));
5     cep = real(ifft(log(abs(X)+eps)));
6     [~, q] = max(cep(30:200));
7     pitch(i) = fs/(q+29);
8 end

```

Explanation: Pitch is estimated from the location of the dominant cepstral peak. The pitch contour represents fundamental frequency variations across time.

Task 9: SNR Evaluation

MATLAB Code:

```

1 snr_before = snr(x_filt);
2 snr_after = snr(enhanced_filt);
3
4 fprintf('SNR Before Enhancement: %.2f dB\n', snr_before);
5 fprintf('SNR After Enhancement : %.2f dB\n', snr_after);

```

Explanation: Signal-to-Noise Ratio (SNR) is computed before and after enhancement to quantitatively measure improvement in speech quality.

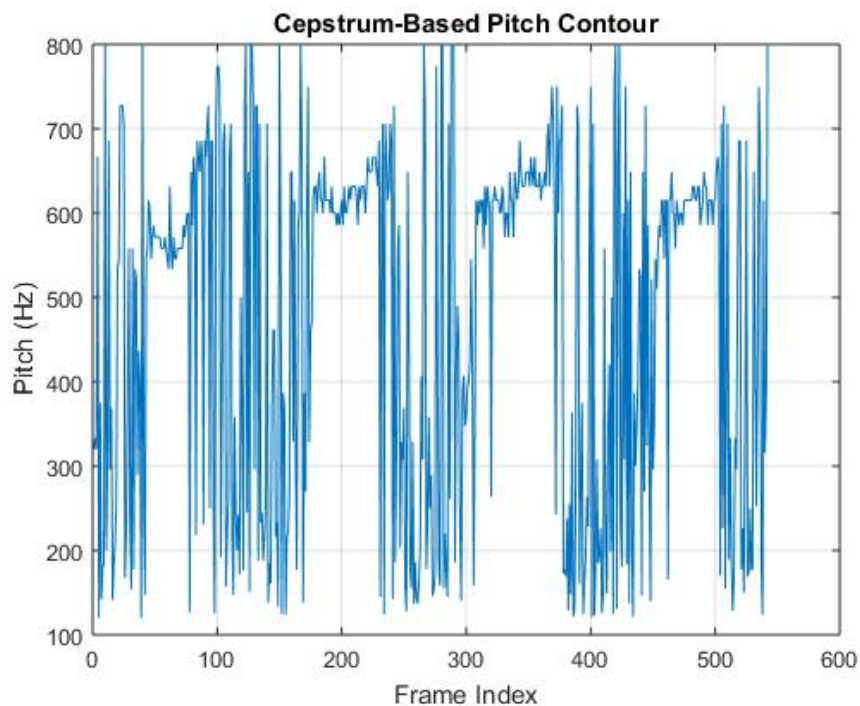


Figure 1: Cepstrum-Based Pitch Contour

Task 10: Spectrogram and Waveform Analysis

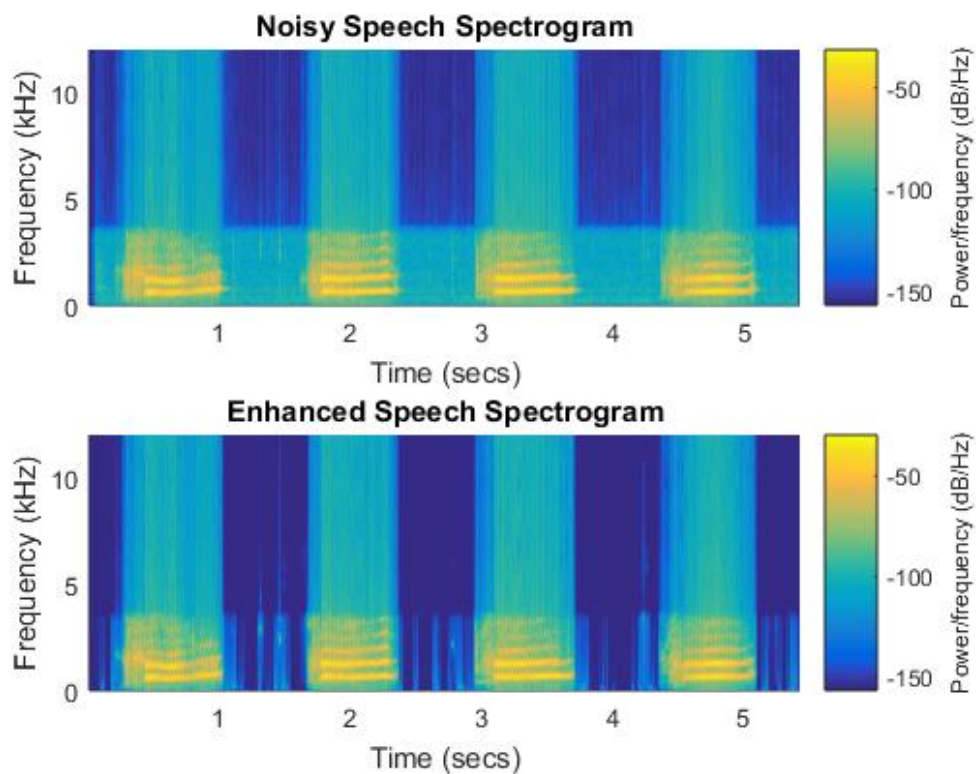


Figure 2: Spectrogram Comparison of Noisy and Enhanced Speech

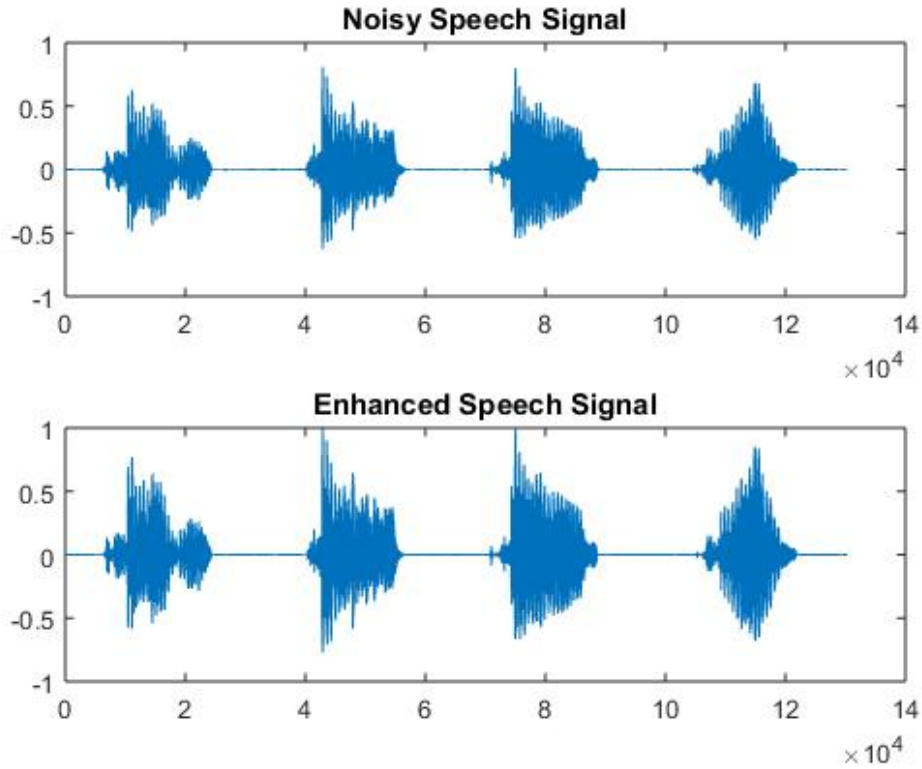


Figure 3: Time-Domain Waveform Comparison

Explanation: Spectrograms show reduced background noise after enhancement, while time-domain plots demonstrate improved waveform smoothness and clarity.

Conclusion

In this open-ended lab, a complete speech enhancement system was successfully implemented using MATLAB. The combination of VAD, noise estimation, spectral subtraction, cepstrum-based pitch detection, and bandpass filtering significantly improved speech quality. Both objective (SNR) and subjective (spectrogram and waveform) analyses confirmed the effectiveness of the proposed approach. This lab enhanced practical understanding of real-world speech signal processing techniques.